

```

Program.cs
App
1 using System;
2 using System.Linq.Expressions;
3 using System.Runtime.InteropServices;
4 using System.Security.Cryptography;
5 using System.Security.Cryptography.X509Certificates;
6 using System.Threading.Tasks;
7 using System.Xml.Linq;
8
9 Ссылка: 0
class Program
10 {
11     Ссылка: 0
12     static void Main()
13     {
14         Console.WriteLine("==== 3.1. Арифметические операторы =====\n");
15         Console.WriteLine("1: x=" + (17 / 5) + ", y=" + (17 % 5));
16         int a1 = 5; Console.WriteLine("2: res=" + (++a1 * 2));
17         int a2 = 5; Console.WriteLine("3: res=" + (a2++ * 2));
18         Console.WriteLine("4: " + (7 / 2) + " и " + (7.0 / 2));
19         Console.WriteLine("5: " + (-15 % 4));
20         int x6 = 10; x6 = x6++ + ++x6; Console.WriteLine("6: " + x6);
21         int max8 = int.MaxValue; Console.WriteLine("8: " + unchecked(max8 + 1));
22         Console.WriteLine("9: " + (1.0 / 0.0) + " и " + (0.0 / 0.0));
23         int a10 = 8, b10 = 3; Console.WriteLine("10: " + (a10 - b10 * 2 + a10 / b10) + "\n");
24
25         Console.WriteLine("==== 3.2. Операторы сравнения и равенства =====\n");
26         Console.WriteLine("1: " + (5 > 3) + ", " + (5 >= 5));
27         Console.WriteLine("2: " + ("hello" == "hello"));
28         Console.WriteLine("3: " + ("double.NaN" == "double.NaN"));
29         object oa = new int[] { 1 }, ob = new int[] { 1 }; Console.WriteLine("4: " + (oa == ob));
30         Console.WriteLine("5: " + (10 != 10.0));
31         Console.WriteLine("6: " + ("null" == "null"));
32         Console.WriteLine("7: " + ((3 < 5) == (10 >= 20)));
33         Console.WriteLine("8: " + (4 <= 4 && 5 > 2));
34         char c9 = 'b'; Console.WriteLine("9: " + (c9 > 'a'));
35         Console.WriteLine("10: " + (-0.0 == 0.0) + "\n");
36
37         Console.WriteLine("==== 3.3. Логические операторы =====\n");
38         Console.WriteLine("1: " + (!true || false && true));
39         Console.WriteLine("2: Foo не вызван (&&)");
40         Console.WriteLine("3: Foo вызван (&)");
41         Console.WriteLine("4: " + (true ^ false ^ true));
42         Console.WriteLine("5: " + (!(5 > 2 || 3 < 1)));
43         bool a6 = true, b6 = false; Console.WriteLine("6: " + (a6 && !b6 || b6 && !a6));
44         int x7 = 5; Console.WriteLine("7: " + (true || (x7 / 0 == 1)));
45         string A = "Де";
46         string B = "Морган";
47         Console.WriteLine($"9: !({A} && {B}) = !{A} || !{B}");
48         Console.WriteLine($"9: !({A} || {B}) = !{A} && !{B} + "\n");
49
50         Console.WriteLine("==== 3.4. Побитовые операторы и сдвиги =====\n");
51         Console.WriteLine("1: " + (5 & 3));
52         Console.WriteLine("2: " + (5 | 3));
53         Console.WriteLine("3: " + (5 ^ 3));
54         Console.WriteLine("4: " + (~0));
55         Console.WriteLine("5: " + (1 << 4));
56         Console.WriteLine("6: " + (40 >> 2));
57         int n7 = 10; Console.WriteLine("7: " + ((n7 & 8) != 0));
58         int n8 = 0; n8 = n8 | (1 << 2); Console.WriteLine("8: " + n8);
59         int n9 = 31; n9 = n9 & ~(1 << 4); Console.WriteLine("9: " + n9);
60         Console.WriteLine("10: " + ((-16) >> 2) + "\n");
61
62         Console.WriteLine("==== 3.5. Операторы присваивания =====\n");
63         int p1 = 10; p1 += 5; Console.WriteLine("1: " + p1);
64

```

```

Консоль отладки Microsoft Vi
===== 35 сложносоставных заданий на логические выражения =====
1: True
2: False
3: True
4: True
5: True
6: False
7: True
8: True
9: True
10: True
11: True
12: True
13: True
14: True
15: True
16: True
17: False
18: True
19: True
20: True
21: True
22: True
23: True
24: True
25: True
26: True
27: True
28: True
29: True
30: False
31: True
32: True
33: False
34: False
35: True

```

```

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2 using System.Linq.Expressions;
3 using System.Runtime.InteropServices;
4 using System.Security.Cryptography;
5 using System.Security.Cryptography.X509Certificates;
6 using System.Threading.Tasks;
7 using System.Xml.Linq;
8
9 Ссылка: 0
10 class Program
11 {
12     Ссылка: 0
13     static void Main()
14     {
15         Console.WriteLine("==== 3.1. Арифметические операторы =====\n");
16         Console.WriteLine("1: x=" + (17 / 5) + ", y=" + (17 % 5));
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20         Console.WriteLine("5: " + (-15 % 4));
21         int x6 = 10; x6 = x6++ + ++x6; Console.WriteLine("6: " + x6);
22         int max8 = int.MaxValue; Console.WriteLine("8: " + unchecked(max8 + 1));
23         Console.WriteLine("9: " + (1.0 / 0.0) + " и " + (0.0 / 0.0));
24         int a10 = 8, b10 = 3; Console.WriteLine("10: " + (a10 - b10 * 2 + a10 / b10) + "\n");
25
26         Console.WriteLine("==== 3.2. Операторы сравнения и равенства =====\n");
27         Console.WriteLine("1: " + (5 > 3) + ", " + (5 >= 5));
28         Console.WriteLine("2: " + ("hello" == "hello"));
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35         char c9 = 'b'; Console.WriteLine("9: " + (c9 > 'a'));
36         Console.WriteLine("10: " + (-0.0 == 0.0) + "\n");
37
38         Console.WriteLine("==== 3.3. Логические операторы =====\n");
39         Console.WriteLine("1: " + (!true || false && true));
40         Console.WriteLine("2: Foo не вызван (&&));");
41         Console.WriteLine("3: Foo вызван (&));");
42         Console.WriteLine("4: " + (true ^ false ^ true));
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44         bool a6 = true, b6 = false; Console.WriteLine("6: " + (a6 && !b6 || b6 && !a6));
45         int x7 = 5; Console.WriteLine("7: " + (true || (x7 / 0 == 1)));
46         string A = "Де";
47         string B = "Морган";
48         Console.WriteLine($"9: !({A} && {B}) = !{A} || !{B}");
49         Console.WriteLine($"9: !({A} || {B}) = !{A} && !{B}" + "\n");
50
51         Console.WriteLine("==== 3.4. Побитовые операторы и сдвиги =====\n");
52         Console.WriteLine("1: " + (5 & 3));
53         Console.WriteLine("2: " + (5 | 3));
54         Console.WriteLine("3: " + (5 ^ 3));
55         Console.WriteLine("4: " + (~0));
56         Console.WriteLine("5: " + (1 << 4));
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60         int n9 = 31; n9 = n9 & ~(1 << 4); Console.WriteLine("9: " + n9);
61         Console.WriteLine("10: " + ((-16) >> 2) + "\n");
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63         Console.WriteLine("==== 3.5. Операторы присваивания =====\n");
64         int p1 = 10; p1 += 5; Console.WriteLine("1: " + p1);

```

==== 3.1. Арифметические операторы =====

```

1: x=3, y=2
2: res=12
3: res=10
4: 3 и 3,5
5: -3
6: 22
8: -2147483648
9: ? и не число
10: 4

```

==== 3.2. Операторы сравнения и равенства =====

```

1: True, True
2: True
3: True
4: False
5: False
6: True
7: False
8: True
9: True
10: True

```

==== 3.3. Логические операторы =====

```

1: False
2: Foo не вызван (&&)
3: Foo вызван (&)
4: False
5: False
6: True
7: True
9: !(Де && Морган) = !Де || !Морган
9: !(Де || Морган) = !Де && !Морган

```

==== 3.4. Побитовые операторы и сдвиги =====

```

1: 1
2: 7
3: 6
4: -1
5: 16
6: 10
7: True
8: 4
9: 15
10: -4

```

==== 3.5. Операторы присваивания =====

```

1: 15
2: 50
3: 3
4: x ??= y - присвоить, если x == null
5: default
6: 3
7: 10
8: 10
9: 3
10: 0

```

```

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62
63         Console.WriteLine("==== 3.5. Операторы присваивания =====\n");
64         int p1 = 10; p1 += 5; Console.WriteLine("1: " + p1);
65
66         Console.WriteLine("==== 3.6. Тернарный и null-операторы =====\n");
67         Console.WriteLine("1: Pass");
68         Console.WriteLine("2: 50");
69         Console.WriteLine("3: System.Double");
70         Console.WriteLine("4:");
71         Console.WriteLine("5: int? (Nullable<int>)");
72         Console.WriteLine("6: Anonymous");
73         Console.WriteLine("7: User");
74         Console.WriteLine("8: 42");
75         Console.WriteLine("9: var x = condition ? 10 : \"text\" - не компилируется");
76         Console.WriteLine("10: -1");
77
78         Console.WriteLine("==== 3.7. Операторы типов и приведения =====\n");
79         Console.WriteLine("1: True");
80         Console.WriteLine("2:");
81         Console.WriteLine("4: True");
82         Console.WriteLine("5: 8");
83         Console.WriteLine("6: False");
84         Console.WriteLine("7: True");
85         Console.WriteLine("8: 3");
86         Console.WriteLine("9: да, val=42");
87         Console.WriteLine("10: 0 и null");
88     }
89 }

```

```

Консоль отладки Micro
===== 3.5. Операторы присваивания =====
1: 15
2: 50
3: 3
4: x ??= y - присвоить, если x == null
5: default
6: 3
7: 10
8: 10
9: 3
10: 0

===== 3.6. Тернарный и null-операторы =====
1: Pass
2: 50
3: System.Double
4:
5: int? (Nullable<int>)
6: Anonymous
7: User
8: 42
9: var x = condition ? 10 : "text" - не компилируется
10: -1

===== 3.7. Операторы типов и приведения =====
1: True
2:
4: True
5: 8
6: False
7: True
8: 3
9: да, val=42
10: 0 и null

```